

2024



Optimizing Price Performance Using AWS Graviton–Based Instances with Sensor Tower

Learn how digital intelligence and analytics company Sensor Tower boosted price performance and fueled innovation using AWS Graviton–based instances.

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Up to 37%

cost reduction for Redis workloads

Up to 20%

cost reduction for MongoDB workloads

Up to 40%

storage-size reduction using MongoDB



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Overview



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[Sensor Tower](#) provides market intelligence and performance metrics for mobile apps and digital advertising. Its products depend on an extremely available, efficient infrastructure to support the high-quality performance and optimal speed to more than 1,300 enterprise clients globally. For computation and storage, Sensor Tower relies on its highly skilled engineers to manage its pipelines and processing engines using Amazon Web Services (AWS).



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Since its inception, Sensor Tower has run on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. In its continual quest for simplification and cost savings, the company migrated some of its workloads to [AWS Graviton processors](#), a family of processors designed to deliver excellent price performance for cloud workloads running in Amazon EC2. Using AWS Graviton-based instances, Sensor Tower cut compute costs by up to 20 percent while improving performance and storage use and gaining corresponding sustainability benefits. Thus, the company continues to offer reliable, data-driven products to customers at increasing speeds and reduced latency.



Opportunity | Using AWS Graviton–Based Instances for Optimal Price Performance for Sensor Tower

Sensor Tower’s algorithms process trillions of aggregated data points from millions of devices, delivering enterprise-grade datasets to customers. “We provide outside-in analytics for the digital economy,” says Stefan Lynggaard, vice president of engineering at Sensor Tower. “We collect and aggregate data into top-level insights so that decision-makers in the digital economy can use them to run their businesses. We want our data to be as live as it can get.”

Unlike many of its competitors, Sensor Tower acquires data from its own sources and does not rely primarily on external datasets. It collects data from sources such as mobile and desktop users, anonymizes and aggregates it, and then sends it through an ingestion pipeline to AWS-hosted databases. Additionally, the company runs various types of algorithms so that the data is transformed for consumption by its customers. Sensor Tower also provides a frontend software as a service through which customers browse data.

In 2019, AWS announced the general preview of AWS Graviton2-based Amazon EC2 instances. Sensor Tower began to investigate migrating some of its backend workflows to these instances, which provide optimal price performance and are optimized for compute, memory, storage, or general-purpose workloads.

Solution | Cutting Compute Costs by up to 20% Using AWS Graviton-Based Instances

As a first step, Sensor Tower’s engineers migrated eight instances that powered Redis, an open-source in-memory data structure that the company uses for backend pipeline processing. They replaced their large Redis instances—which use 16 CPUs with 128 GB of RAM—with similarly sized [Amazon EC2 R6g Instances](#), which provide high price performance for memory-intensive workloads in Amazon EC2. In addition to reducing monthly costs by up to 37 percent, the company also unexpectedly experienced CPU load decreases across all Redis instances. “It was a pure win,” says Eugene Kryudziuk, senior engineering manager at Sensor Tower. “We cut costs while enhancing and accelerating our performance.”

Buoyed by the success of the Redis migration, the engineers looked to migrate the 40 instances that power critical workloads on MongoDB, a document database designed for ease of application development and scaling. The company saved up to 20 percent on direct compute costs in addition to uncovering indirect cost benefits. For example, the engineers could switch to a more efficient MongoDB compression algorithm, reducing the storage size by up to 40 percent and increasing the speed of data access. “Compute is compute,” says Lynggaard. “Why would you buy more expensive compute if you only care about throughput? It’s a no-brainer if you can switch to something with better performance per dollar.”

Sensor Tower’s cost reduction has translated into a decrease in power consumption. Using AWS Graviton-based instances, the company saves up to 60 percent on energy. “When we consume less compute, we burn fewer fossil fuels and use less energy,” says Lynggaard. “So, whenever we save a dollar, that translates directly into environmental impact. It’s one of our big values at Sensor Tower to act ethically. We try to be a really good citizen of the global economy.”

Through its sophisticated self-managed infrastructure, large databases, and culture of innovation, Sensor Tower attracts top engineering talent. Its engineers continue to adopt cutting-edge generations of processors, migrating to AWS Graviton3-based instances in late 2022 and awaiting the general release of [AWS Graviton4](#).



instances, which will increase the compute performance, number of cores, and memory bandwidth. “We are very fortunate to be able to spin up as much hardware as we want on AWS on very short notice,” says Lynggaard. “Using AWS, we can innovate because we don’t have to set up physical servers to experiment.”

Outcome | Reinvesting Cost Savings into Research and Development

Next, Sensor Tower plans to migrate the servers coded in the general-purpose programming language Ruby, which it uses to run background data processing. The company’s engineers have begun to compile the libraries attached to Ruby servers for ARM so that they can implement cutting-edge AWS Graviton-based instances.

Meanwhile, the company is reinvesting much of its cost savings into research and development so that it can continue to improve its tools’ interaction speed for customers and accelerate new product development. “It’s key to our innovation that AWS Graviton-based instances are well-tested, mature instance types that we can get off the shelf,” says Lynggaard. “Rather than deal with hardware, we can immediately focus on our products and serve our users.”

About Sensor Tower

Based in San Francisco, Sensor Tower focuses on providing enterprise-grade market intelligence and performance metrics in the realms of mobile apps and digital advertising.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS Graviton Processors

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

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